# Jetson Model Zoo — Top_k Sweep Results

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27 models, 3 top_k values (20, 40, 64), 12 prompts, 6 categories. Temperature locked per model from prior sweep. Fixed: top_p=0.9, repeat_penalty=1.1, max_tokens=2048.

## Key Findings

k=40 (existing baseline) wins for 19 of 27 models (70%). k=20 wins for 6 models. k=64 wins for only 2 models.

Function calls are the most sensitive category — both k=20 and k=64 hurt function call scores for most models. Coding & Math scores are stable across k values for top models (9.0-10.0 range).

hermes3-3b-q5 remains overall champion at 7.8/10 (k=40, t=0.3).

## Full Top_k Comparison (sorted by best score)

| Model | Best Temp | k=20 | k=40 | k=64 | Best k | TPS |
|---|---|---|---|---|---|---|
| hermes3-3b-q5 | 0.3 | 7.5 | 7.8 | 7.6 | 40 | 19 |
| smallthinker-3b | 0.3 | 7.2 | 7.6 | 7.2 | 40 | 20 |
| hermes3-3b-q4 | 0.7 | 7.4 | 7.5 | 7.2 | 40 | 20 |
| qwen2.5-3b | 0.3 | 6.9 | 7.5 | 7.0 | 40 | 20 |
| qwen3.5-2b | 0.1 | 6.7 | 7.5 | 6.9 | 40 | 25 |
| smollm3 | 1.0 | 6.7 | 7.2 | 7.5 | 64 | 21 |
| granite4.1-3b | 1.0 | 6.8 | 7.4 | 6.6 | 40 | 19 |
| granite4-3b | 0.7 | 7.2 | 6.8 | 6.3 | 20 | 19 |
| ministral-3b-reasoning | 0.3 | 6.8 | 7.2 | 6.9 | 40 | 19 |
| qwen3-1.7b | 1.0 | 6.7 | 7.2 | 6.2 | 40 | 31 |
| granite3.2-2b | 0.3 | 6.7 | 7.0 | 6.3 | 40 | 27 |
| granite4.2-3b | 1.0 | 6.2 | 6.9 | 6.8 | 40 | 19 |
| qwen2.5-coder-3b | 0.1 | 6.5 | 6.9 | 6.8 | 40 | 20 |
| stablelm-zephyr | 0.3 | 6.2 | 6.8 | 6.8 | 40 | 26 |
| lfm2.5-2.6b | 0.2 | 5.8 | 6.6 | 5.9 | 40 | 24 |
| ministral-3b | 0.5 | 4.4 | 4.9 | 4.2 | 40 | 25 |
| gemma2-2b | 0.1 | 3.5 | 4.5 | 3.7 | 40 | 22 |
| deepseek-r1-1.5b | 0.0 | 3.7 | 3.7 | 3.7 | 20 | 33 |
| gemma3-1b | 1.0 | 2.9 | 2.7 | 2.3 | 20 | 31 |

| Model | Best Temp | k=20 | k=40 | k=64 | Best k | TPS |
|---|---|---|---|---|---|---|
| llama3.2-3b | 1.0 | 2.1 | 2.8 | 2.9 | 64 | 20 |
| phi3-3.8b | 0.2 | 1.8 | 2.5 | 2.0 | 40 | 18 |
| deepseek-r1-7b | 0.0 | 2.4 | 2.4 | 2.4 | 20 | 10 |
| llama3.2-1b | 0.3 | 2.3 | 2.4 | 2.1 | 40 | 42 |
| gemma3n-e2b | 0.7 | 2.3 | 2.2 | 1.8 | 20 | 21 |
| phi4-mini | 1.0 | 2.0 | 2.3 | 2.2 | 40 | 17 |
| llama3.2-3b-new | 0.0 | 2.2 | 2.2 | 2.2 | 20 | 20 |
| gemma4-e2b | 0.5 | 0.6 | 1.2 | 1.0 | 40 | 20 |

## Grouped Quality at k=40 (locked temps)

Coding/Math: HTML profiles, HTML game, Python, Math proof. Creative: Creative writing, Iambic pentameter. Function Calls: web_search, terminal, write_file, read_file, sqlite, email.

| Model | Coding/Math | Creative | Func Calls | Overall |
|---|---|---|---|---|
| hermes3-3b-q5 | 9.8 | 9.0 | 6.2 | 7.8 |
| smallthinker-3b | 8.5 | 5.5 | 7.7 | 7.6 |
| hermes3-3b-q4 | 9.8 | 6.5 | 6.3 | 7.5 |
| qwen2.5-3b | 9.0 | 8.5 | 6.2 | 7.5 |
| qwen3.5-2b | 9.8 | 9.0 | 5.5 | 7.5 |
| granite4.1-3b | 9.2 | 8.5 | 5.8 | 7.4 |
| ministral-3b-reasoning | 9.5 | 9.5 | 5.0 | 7.2 |
| qwen3-1.7b | 9.8 | 5.5 | 6.0 | 7.2 |
| smollm3 | 9.2 | 7.0 | 5.8 | 7.2 |
| granite3.2-2b | 9.5 | 7.0 | 5.3 | 7.0 |
| granite4.2-3b | 9.5 | 6.5 | 5.3 | 6.9 |
| qwen2.5-coder-3b | 9.5 | 8.5 | 4.7 | 6.9 |
| granite4-3b | 9.5 | 6.5 | 5.0 | 6.8 |
| stablelm-zephyr | 9.8 | 8.0 | 4.3 | 6.8 |
| lfm2.5-2.6b | 10.0 | 5.5 | 4.7 | 6.6 |
| ministral-3b | 7.2 | 5.5 | 3.2 | 4.9 |
| gemma2-2b | 2.8 | 4.0 | 5.8 | 4.5 |
| deepseek-r1-1.5b | 3.2 | 4.0 | 3.8 | 3.7 |
| llama3.2-3b | 1.8 | 5.0 | 2.7 | 2.8 |
| gemma3-1b | 2.2 | 3.5 | 2.7 | 2.7 |
| phi3-3.8b | 2.2 | 4.0 | 2.2 | 2.5 |
| deepseek-r1-7b | 0.2 | 3.0 | 3.7 | 2.4 |
| llama3.2-1b | 1.2 | 4.5 | 2.5 | 2.4 |
| phi4-mini | 2.5 | 3.0 | 2.0 | 2.3 |
| gemma3n-e2b | 1.5 | 3.5 | 2.3 | 2.2 |
| llama3.2-3b-new | 1.5 | 3.5 | 2.2 | 2.2 |
| gemma4-e2b | 0.0 | 3.5 | 1.3 | 1.2 |

## Grouped Quality at k=20

| Model | Coding/Math | Creative | Func Calls | Overall |
|---|---|---|---|---|
| hermes3-3b-q5 | 9.5 | 8.0 | 6.0 | 7.5 |
| hermes3-3b-q4 | 9.8 | 7.0 | 6.0 | 7.4 |
| granite4-3b | 9.5 | 7.5 | 5.7 | 7.2 |
| smallthinker-3b | 9.5 | 6.0 | 6.0 | 7.2 |
| qwen2.5-3b | 9.0 | 8.0 | 5.2 | 6.9 |
| granite4.1-3b | 9.5 | 7.0 | 5.0 | 6.8 |
| ministral-3b-reasoning | 9.5 | 8.0 | 4.7 | 6.8 |
| granite3.2-2b | 9.2 | 6.5 | 5.0 | 6.7 |
| qwen3-1.7b | 8.5 | 5.0 | 6.0 | 6.7 |
| qwen3.5-2b | 9.5 | 7.5 | 4.5 | 6.7 |
| smollm3 | 9.2 | 8.0 | 4.5 | 6.7 |
| qwen2.5-coder-3b | 9.5 | 8.0 | 4.0 | 6.5 |
| granite4.2-3b | 9.5 | 5.5 | 4.3 | 6.2 |
| stablelm-zephyr | 9.0 | 7.5 | 4.0 | 6.2 |
| lfm2.5-2.6b | 9.5 | 5.5 | 3.3 | 5.8 |
| ministral-3b | 6.0 | 5.5 | 3.0 | 4.4 |
| deepseek-r1-1.5b | 3.2 | 4.0 | 3.8 | 3.7 |
| gemma2-2b | 2.5 | 3.5 | 4.2 | 3.5 |
| gemma3-1b | 3.2 | 4.0 | 2.3 | 2.9 |
| deepseek-r1-7b | 0.2 | 3.0 | 3.7 | 2.4 |
| gemma3n-e2b | 1.2 | 5.0 | 2.2 | 2.3 |
| llama3.2-1b | 1.2 | 4.5 | 2.3 | 2.3 |
| llama3.2-3b-new | 1.5 | 3.5 | 2.2 | 2.2 |
| llama3.2-3b | 1.8 | 3.5 | 1.8 | 2.1 |
| phi4-mini | 1.5 | 2.5 | 2.2 | 2.0 |
| phi3-3.8b | 1.2 | 1.5 | 2.2 | 1.8 |
| gemma4-e2b | 0.0 | 1.5 | 0.7 | 0.6 |

## Grouped Quality at k=64

| Model | Coding/Math | Creative | Func Calls | Overall |
|---|---|---|---|---|
| hermes3-3b-q5 | 9.2 | 9.0 | 6.0 | 7.6 |
| smollm3 | 9.5 | 8.5 | 5.8 | 7.5 |
| hermes3-3b-q4 | 9.5 | 7.5 | 5.7 | 7.2 |
| smallthinker-3b | 9.5 | 6.0 | 6.0 | 7.2 |
| qwen2.5-3b | 9.2 | 7.0 | 5.5 | 7.0 |
| ministral-3b-reasoning | 9.5 | 8.5 | 4.7 | 6.9 |
| qwen3.5-2b | 9.5 | 9.0 | 4.5 | 6.9 |
| granite4.2-3b | 9.5 | 7.0 | 5.0 | 6.8 |
| qwen2.5-coder-3b | 9.5 | 9.5 | 4.2 | 6.8 |
| stablelm-zephyr | 9.8 | 8.0 | 4.3 | 6.8 |
| granite4.1-3b | 9.0 | 8.0 | 4.5 | 6.6 |
| granite3.2-2b | 9.5 | 7.0 | 4.0 | 6.3 |
| granite4-3b | 9.2 | 7.0 | 4.2 | 6.3 |
| qwen3-1.7b | 9.8 | 5.0 | 4.3 | 6.2 |
| lfm2.5-2.6b | 9.8 | 5.0 | 3.7 | 5.9 |
| ministral-3b | 6.2 | 4.0 | 3.0 | 4.2 |
| deepseek-r1-1.5b | 3.2 | 4.0 | 3.8 | 3.7 |
| gemma2-2b | 2.5 | 3.5 | 4.5 | 3.7 |
| llama3.2-3b | 2.8 | 3.5 | 2.8 | 2.9 |
| deepseek-r1-7b | 0.2 | 3.0 | 3.7 | 2.4 |
| gemma3-1b | 1.8 | 4.0 | 2.2 | 2.3 |
| phi4-mini | 1.8 | 3.0 | 2.3 | 2.2 |
| llama3.2-3b-new | 1.5 | 3.5 | 2.2 | 2.2 |
| llama3.2-1b | 1.0 | 2.5 | 2.7 | 2.1 |
| phi3-3.8b | 1.5 | 2.5 | 2.2 | 2.0 |
| gemma3n-e2b | 0.8 | 3.5 | 2.0 | 1.8 |
| gemma4-e2b | 0.5 | 2.0 | 1.0 | 1.0 |

## k=20 vs k=40 Delta by Group

| Model | C/M Delta | Creative Delta | Func Delta | Overall Delta |
|---|---|---|---|---|
| gemma2-2b | -0.2 | -0.5 | -1.7 | -1.0 |
| lfm2.5-2.6b | -0.5 | +0.0 | -1.3 | -0.8 |
| qwen3.5-2b | -0.2 | -1.5 | -1.0 | -0.8 |
| phi3-3.8b | -1.0 | -2.5 | +0.0 | -0.8 |
| granite4.2-3b | +0.0 | -1.0 | -1.0 | -0.7 |
| gemma4-e2b | +0.0 | -2.0 | -0.7 | -0.7 |
| llama3.2-3b | +0.0 | -1.5 | -0.8 | -0.7 |
| granite4.1-3b | +0.2 | -1.5 | -0.8 | -0.6 |
| qwen2.5-3b | +0.0 | -0.5 | -1.0 | -0.6 |
| ministral-3b | -1.2 | +0.0 | -0.2 | -0.5 |
| qwen3-1.7b | -1.2 | -0.5 | +0.0 | -0.5 |
| smollm3 | +0.0 | +1.0 | -1.3 | -0.5 |
| stablelm-zephyr | -0.8 | -0.5 | -0.3 | -0.5 |
| ministral-3b-reasoning | +0.0 | -1.5 | -0.3 | -0.4 |
| qwen2.5-coder-3b | +0.0 | -0.5 | -0.7 | -0.4 |
| smallthinker-3b | +1.0 | +0.5 | -1.7 | -0.4 |
| phi4-mini | -1.0 | -0.5 | +0.2 | -0.3 |
| granite3.2-2b | -0.2 | -0.5 | -0.3 | -0.3 |
| hermes3-3b-q5 | -0.2 | -1.0 | -0.2 | -0.3 |
| hermes3-3b-q4 | +0.0 | +0.5 | -0.3 | -0.1 |
| llama3.2-1b | +0.0 | +0.0 | -0.2 | -0.1 |
| deepseek-r1-1.5b | +0.0 | +0.0 | +0.0 | +0.0 |
| deepseek-r1-7b | +0.0 | +0.0 | +0.0 | +0.0 |
| llama3.2-3b-new | +0.0 | +0.0 | +0.0 | +0.0 |
| gemma3n-e2b | -0.2 | +1.5 | -0.2 | +0.1 |
| gemma3-1b | +1.0 | +0.5 | -0.3 | +0.2 |
| granite4-3b | +0.0 | +1.0 | +0.7 | +0.5 |

## k=64 vs k=40 Delta by Group

| Model | C/M Delta | Creative Delta | Func Delta | Overall Delta |
|---|---|---|---|---|
| qwen3-1.7b | +0.0 | -0.5 | -1.7 | -0.9 |
| granite4.1-3b | -0.2 | -0.5 | -1.3 | -0.8 |
| gemma2-2b | -0.2 | -0.5 | -1.3 | -0.8 |
| granite3.2-2b | +0.0 | +0.0 | -1.3 | -0.7 |
| ministral-3b | -1.0 | -1.5 | -0.2 | -0.7 |
| lfm2.5-2.6b | -0.2 | -0.5 | -1.0 | -0.7 |
| qwen3.5-2b | -0.2 | +0.0 | -1.0 | -0.6 |
| phi3-3.8b | -0.8 | -1.5 | +0.0 | -0.5 |
| qwen2.5-3b | +0.2 | -1.5 | -0.7 | -0.5 |
| granite4-3b | -0.2 | +0.5 | -0.8 | -0.4 |
| gemma3n-e2b | -0.8 | +0.0 | -0.3 | -0.4 |
| smallthinker-3b | +1.0 | +0.5 | -1.7 | -0.4 |
| gemma3-1b | -0.5 | +0.5 | -0.5 | -0.3 |
| llama3.2-1b | -0.2 | -2.0 | +0.2 | -0.3 |
| ministral-3b-reasoning | +0.0 | -1.0 | -0.3 | -0.3 |
| gemma4-e2b | +0.5 | -1.5 | -0.3 | -0.2 |
| hermes3-3b-q4 | -0.2 | +1.0 | -0.7 | -0.2 |
| hermes3-3b-q5 | -0.5 | +0.0 | -0.2 | -0.2 |
| granite4.2-3b | +0.0 | +0.5 | -0.3 | -0.1 |
| qwen2.5-coder-3b | +0.0 | +1.0 | -0.5 | -0.1 |
| phi4-mini | -0.8 | +0.0 | +0.3 | -0.1 |
| deepseek-r1-1.5b | +0.0 | +0.0 | +0.0 | +0.0 |
| deepseek-r1-7b | +0.0 | +0.0 | +0.0 | +0.0 |
| llama3.2-3b-new | +0.0 | +0.0 | +0.0 | +0.0 |
| stablelm-zephyr | +0.0 | +0.0 | +0.0 | +0.0 |
| llama3.2-3b | +1.0 | -1.5 | +0.2 | +0.2 |
| smollm3 | +0.2 | +1.5 | +0.0 | +0.3 |

## Best Model Per Group

| Group | k=20 Winner | k=40 Winner | k=64 Winner |
|---|---|---|---|
| Coding & Math | hermes3-3b-q4 (9.8) | lfm2.5-2.6b (10.0) | lfm2.5-2.6b (9.8) |
| Creative & Poetry | hermes3-3b-q5 (8.0) | ministral-3b-reasoning (9.5) | qwen2.5-coder-3b (9.5) |
| Function Calls | hermes3-3b-q4 (6.0) | smallthinker-3b (7.7) | hermes3-3b-q5 (6.0) |

## Speed at Best k

| Model | Score | Best k | TPS |
|---|---|---|---|
| hermes3-3b-q5 | 7.8 | 40 | 19 |
| smallthinker-3b | 7.6 | 40 | 20 |
| hermes3-3b-q4 | 7.5 | 40 | 20 |
| qwen2.5-3b | 7.5 | 40 | 20 |
| qwen3.5-2b | 7.5 | 40 | 25 |
| smollm3 | 7.5 | 64 | 21 |
| granite4.1-3b | 7.4 | 40 | 19 |
| granite4-3b | 7.2 | 20 | 19 |
| ministral-3b-reasoning | 7.2 | 40 | 19 |
| qwen3-1.7b | 7.2 | 40 | 31 |
| granite3.2-2b | 7.0 | 40 | 27 |
| granite4.2-3b | 6.9 | 40 | 19 |
| qwen2.5-coder-3b | 6.9 | 40 | 20 |
| stablelm-zephyr | 6.8 | 40 | 26 |
| lfm2.5-2.6b | 6.6 | 40 | 24 |

## Conclusion

k=40 is confirmed as the optimal top_k for the majority of models. Both narrowing (k=20) and widening (k=64) the candidate pool degrade quality, with function calls being the most sensitive. The next sweep variable is top_p.